

SHLOK MISHRA

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PROFESSIONAL SUMMARY

Hands-on experience deploying two institutional platforms and building explainable machine-learning, computer-vision, and LLM/RAG systems through Python-based development workflows.

EDUCATION

G.H. Raisoni College of Engineering, Nagpur

2023 – 2027

Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence)

Nutan Vidyalay, Gondia

2023

Higher Secondary Certificate (Class XII), Maharashtra State Board (MSBSHSE) – 73.83%

Vivek Mandir, Gondia

2021

Secondary School Certificate (Class X), Central Board of Secondary Education (CBSE) – 70.33%

PROFESSIONAL EXPERIENCE

WORKMATES

May 2026 – August 2026

Software Development Intern | Affiliated with Pragatishil Bahuuddeshiya Sanstha, Washim

- Delivered two production institutional platforms for Sunita Nursing School and Narendra Suryawanshi College of Pharmacy, converting fragmented admissions, academic, faculty, approval, compliance, and contact information into structured digital systems.
- Designed reusable modules and a lightweight internal workflow for institutional records, document access, notices, enquiries, and recurring content updates based on stakeholder requirements.
- Managed production releases through Git/GitHub and Hostinger, including responsive testing, EmailJS integration, DNS/SSL setup, indexing, sitemap and canonical metadata, troubleshooting, and post-launch maintenance.

Live Deployments: [Sunita Nursing School](#) | [Narendra Suryawanshi College of Pharmacy](#)

SELECTED PROJECTS

Prithvi Agro AI — Explainable Agricultural Decision-Support System | [Repository](#)

2025 – 2026

Tech: Python | scikit-learn | Random Forest | XGBoost | LightGBM | SHAP | LLM/RAG | Flask | REST APIs

- Built a bilingual recommendation system combining soil, seasonal weather, crop requirements, and market/MSP data; compared Random Forest, XGBoost, and LightGBM models and added SHAP explanations.
- Developed an LLM/RAG assistant over a structured crop knowledge base to answer farmer questions on cultivation, inputs, risks, and recommended actions in accessible language.

Advanced Deepfake Detection — Explainable Media Forensics & Authenticity Verification

2026

Tech: Python | TensorFlow/Keras | OpenCV | RetinaFace | CNN | Grad-CAM | FastAPI | Web3 Hash Verification

- Created an end-to-end image/video-frame pipeline using RetinaFace extraction, preprocessing, CNN inference, confidence-based classification, and an UNCERTAIN outcome for low-confidence predictions.
- Generated Grad-CAM heatmaps for explainability and built asynchronous FastAPI prediction/verification endpoints with media hashing and blockchain-compatible authenticity checks.

Neuro Detect AI — Brain Tumor MRI Screening Prototype

2025

Tech: Python | TensorFlow/Keras | CNN | OpenCV | NumPy | Image Preprocessing | Automated Reporting

- Trained a CNN-based MRI classifier using resizing, normalization, augmentation, and structured validation, achieving approximately 90% validation accuracy on the project dataset.
- Built a reusable inference and automated-reporting workflow that applies consistent preprocessing, returns confidence scores, and summarizes screening results for demonstration.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: scikit-learn, Random Forest, XGBoost, LightGBM, feature engineering, model evaluation, hyperparameter tuning, SHAP

Deep Learning & Vision: TensorFlow/Keras, CNNs, OpenCV, RetinaFace, Grad-CAM, image preprocessing, inference pipelines

Generative AI: Large Language Models, Retrieval-Augmented Generation, LangChain, LangGraph, prompt engineering, structured knowledge bases, tool calling

Backend & Data: FastAPI, Flask, REST APIs, asynchronous endpoints, Pandas, NumPy, Matplotlib, exploratory data analysis

Development & Deployment: Git, GitHub, VS Code, Jupyter, Google Colab, Vite, Hostinger, EmailJS, Search Console, Chrome DevTools, DNS/SSL, SEO

LEADERSHIP & ACTIVITIES

- Smart India Hackathon Finalist** — Developed an AI-based smart-farming solution using government and geospatial datasets; contributed to model workflow, integration, and presentation.
- IEEE Student Branch Team Lead** — Coordinated technical teams, delegated work, and supported hackathons, workshops, and structured execution plans.
- Google Solution Challenge 2024** — Contributed to a UN SDG-aligned student solution through problem framing, modular application design, Google-supported technologies, and presentation.

SELECTED CERTIFICATIONS

Google Solution Challenge 2026 - Build with AI (2026) — Problem framing and Google-technology-enabled solution development.

Samsung Solve for Tomorrow 2026 (2026) — Technology-led social-impact ideation, feasibility, scalability, and implementation planning.

Google AI Essentials Specialization (January 2026) — Generative AI tools, prompt design, responsible AI, productivity, and output evaluation.

AI Fluency: Framework & Foundations (June 2026) — Anthropic's 4D framework for effective and responsible human-AI collaboration.